

Aptitudinal Alignment

Alignment.V1

Complete Beacon Scoring Engine Baseline plus Universal Brain Architecture

Aptitude Systems — Internal Documentation

Version 1.0 August 2026

The complete Beacon specification is preserved first; newer Alignment layers are additive.

Contents

1	Introduction	4
2	Product Language	4
3	Design Principles	4
4	Canonical Vocabulary	4
4.1	Dimension Symbols	5
5	Evidence and Provenance Model	5
5.1	Source Type Allowlist	5
5.2	Provenance Fields	6
6	Legal Intelligence Pipeline	6
6.1	Flow	6
6.2	Agents	6
6.3	Invariants	7
7	Aptitudinal Alignment Scoring	7
7.1	Flow	7
7.2	Universal Entity Contract	7
7.3	Default Weights	8
7.4	Classification Tiers	8
7.5	Evidence Thresholds and Insufficient Data	8
7.6	Hard Floors	8
7.7	Normalization	9
7.8	Reproducibility	9

8	Vertical Taxonomy	9
8.1	Top-Level Families	9
8.2	First Validation Verticals	9
9	Vertical Profile Contract	10
9.1	Profile Shape	10
9.2	Entity and Activity Separation	11
10	Nationwide Consumer Reporting Agency Profile	11
10.1	Class Definition	11
10.2	Metric Registry	11
10.3	Complete Evidence Inventory	12
10.4	Equifax Pilot Scope	12
11	Department of Corrections Profile	12
11.1	Profile Scope	12
11.2	Candidate Metric Families	13
11.3	DOC-Specific Safeguards	13
12	Executable Metric Registry and Cross-Vertical Validation	13
13	Metric Observation Layer	14
14	Immutable Source Snapshot Layer	14
15	Governance and Reproducibility	15
16	Implementation Phases	15

Introduction

Aptitudinal Alignment measures how closely a public institution's documented conduct aligns with what that institution is capable of and obligated to achieve. It is designed as a reverse-FICO system for public entities: instead of scoring an individual's credit risk, every eligible *entity* — a judge, court, agency, office, or institution — receives one explainable *Alignment Score* derived from the public record.

Each public record, decision, or procedural act is treated as a discrete unit of institutional behavior. These units interact through measurable fields of fidelity, consistency, transparency, efficiency, equity, and accountability. The emergent field defines the Alignment Score.

Product Language

- The page and system name is **Aptitudinal Alignment**.
- The single entity result is the **Alignment Score**.
- The six explainable components are **dimension scores**.
- The interpreted category is the **Alignment Status**.
- Each scored subject is an **entity** with a versioned **potential profile**.

The term “Composite Score” is not a product term. Wherever source material said “Aptitude Score” or “composite,” the unified manual says “Alignment Score.”

Design Principles

- The quantum model is an analytical metaphor and data abstraction — not literal quantum computation.
- Public institutional records are the primary and sufficient evidentiary basis.
- A score is never invented when evidence thresholds are not met; the result is `insufficient_data` with explicit reasons.
- Raw inputs and derived outputs are preserved separately so every score can be recomputed.
- Legal-intelligence conclusions and institutional scores are separate bounded products.
- No Alignment Score is a finding of individual wrongdoing, a prediction of future conduct, or a substitute for an official finding.

Canonical Vocabulary

One vocabulary is used across the codebase, the API, and this manual.

Term	Meaning
Evidence item	A sourced institutional event, record, decision, filing, or report.
Source record	Citation and provenance metadata for an evidence item: source type, credibility tier, date, jurisdiction, and retrieval details.
Behavioral quantum	The normalized analytical representation of an evidence item. It is a data abstraction, not a physical particle.
Metric observation	A raw measurement mapped to a dimension and sub-metric.
Dimension score	A reproducible score for Fidelity, Consistency, Transparency, Efficiency, Equity, or Accountability.
Alignment Score	The universal FICO-like result for an entity: a weighted, explainable measure of how closely documented conduct approaches that entity's defined potential after evidence and hard-floor checks.
Entity	Any subject evaluated by the system: judge, court, agency, office, or other institution.
Potential profile	The versioned standard for what an entity can and should achieve, based on mandate, role, jurisdiction, peer class, resources, and public obligations.
Alignment Status	The interpreted category of the Alignment Score: <code>high_alignment</code> , <code>adequate</code> , <code>at_risk</code> , <code>low_alignment</code> , <code>critical_failure</code> , <code>insufficient_data</code> , or <code>suppressed</code> .

Dimension Symbols

The six dimensions use unambiguous symbols in code and documentation: *F* Fidelity, *C* Consistency, *T* Transparency, *E* Efficiency, *Q* Equity, and *Ac* Accountability. *Q* means Equity; *Ac* means Accountability. No other meaning is attached to these symbols.

Evidence and Provenance Model

Both the legal-intelligence pipeline and the Alignment scoring pipeline use one shared evidence foundation. Each evidence item identifies the institution or actor, the event, the source record, source type and credibility tier, citation, document date, jurisdiction, dimension and sub-metric mapping, raw and normalized values, uncertainty, and the processing, ontology, metric registry, and normalization versions that produced it.

Raw evidence is immutable. Derived observations and scores are append-only revisions linked to the input hashes and configuration versions used to produce them.

Source Type Allowlist

Only records with these `source_type` values are eligible for institutional score computation:

```
public_record, regulatory_filing, court_decision, agency_report,
cfpb_complaint, ftc_action, inspector_general,
individual_submission
```

Records with `source_type = individual_submission` are supplementary (credibility tier 6, weight 0.05) and cannot dominate any dimension unless corroborated by a public record. Their absence never degrades institutional score validity.

Provenance Fields

```
{
  "id": "hash(entityId + eventId + sourceRecordId)",
  "entity_id": "uuid",
  "event_id": "uuid",
  "source_record_id": "uuid",
  "source_type": "court_decision",
  "credibility_tier": 1,
  "citation": "State v. Example, 123 Or App 45 (2024)",
  "document_date": "2024-03-01",
  "jurisdiction_code": "OR",
  "dimension": "F",
  "raw_value": 0.91,
  "normalized_value": 0.91,
  "confidence": 1.0,
  "processing_version": "brain-v1",
  "ontology_version": "ontology-v1",
  "metric_registry_version": "metrics-v1",
  "normalization_version": "norm-v1",
  "input_hash": "hex",
  "recorded_at": "ISO-8601"
}
```

Legal Intelligence Pipeline

The legal-intelligence pipeline transforms lived legal experiences into structured, queryable, auditable legal intelligence. It is a separate bounded product from institutional scoring.

Flow

```
narrative -> structured event -> claims/doctrines
          -> authority graph and retrieval -> procedural pathways
```

Agents

- **Intake Parser:** structures observable facts only; never infers legal conclusions.
- **Legal Classifier:** maps to claims and doctrines using the ontology only; conservative classification.
- **Graph Seeder:** builds the legal knowledge graph with typed edges (supports, governed_by, codified_in, violates, resolved_by).
- **Vector Retrieval:** returns ranked supporting authority with metadata — supporting material only, no conclusions.

- **Procedural Engine:** builds ordered enforcement pathways with prerequisites and jurisdiction alignment.

Invariants

- No output without provenance.
- No citation without source.
- No authority without jurisdiction.
- No overwrite of historical records.
- No legal conclusion without supporting authorities.
- An incident never silently becomes an institutional score.

Aptitudinal Alignment Scoring

Flow

```
entity + potential profile + evidence
-> metric observations
-> normalization
-> six dimension evaluators
-> hard floors
-> Alignment Score
-> tier / status
```

Every supported entity receives the same explainable score contract, but not the same metric registry. A judge, court, agency, prosecutor's office, or credit reporting agency gets a potential profile appropriate to its mandate, role, jurisdiction, peer class, resources, and public obligations.

Universal Entity Contract

```
{
  "entity_id": "uuid",
  "entity_type": "judge | court | agency | office | institution",
  "potential_profile_version": "string",
  "alignment_score": 0.0,
  "alignment_status": "high_alignment | adequate | at_risk |
    low_alignment | critical_failure | insufficient_data |
    suppressed",
  "dimensions": { "F": 0.0, "C": 0.0, "T": 0.0, "E": 0.0, "Q": 0.0,
    "Ac": 0.0 },
  "evidence_summary": { "count": 0, "source_types": [], "
    missing_requirements": [] },
  "flags": [],
  "audit_reference": "string"
}
```

The numeric range is normalized to $[0, 1]$ internally. A FICO-like integer scale may be used only through a documented presentation transform; it must never hide the six dimensions, evidence sufficiency, or limitations.

Default Weights

$$A = \sum_{i=1}^6 w_i \cdot D_i \quad \sum_{i=1}^6 w_i = 1.0$$

Dimension	Symbol	Default weight	Rationale
Fidelity	<i>F</i>	0.22	Legal mandate adherence is foundational
Consistency	<i>C</i>	0.18	Variance in outcomes signals systemic bias
Transparency	<i>T</i>	0.17	Public record access is a legal right
Efficiency	<i>E</i>	0.15	Resource stewardship proxies competence
Equity	<i>Q</i>	0.16	Demographic fairness is a constitutional floor
Accountability	<i>Ac</i>	0.12	Self-correction capacity determines trajectory

Classification Tiers

Tier	Score range	Alignment Status
I	0.85 – 1.00	high_alignment
II	0.70 – 0.84	adequate
III	0.50 – 0.69	at_risk
IV	0.30 – 0.49	low_alignment
V	0.00 – 0.29	critical_failure

Evidence Thresholds and Insufficient Data

A score is emitted only when the configured evidence thresholds pass: at least 12 behavioral quanta per dimension from at least 3 distinct source types. When any dimension fails its threshold, the result is `insufficient_data`: `alignment_score` is null, the failing dimensions are null, and `evidence_summary.missing_requirements` explains why. The system never estimates a missing dimension.

Hard Floors

Equity hard floors override the weighted average and cannot be bypassed:

- Finding of intentional discrimination within 10 years $\Rightarrow Q_{\min} = 0.15$
- Active consent decree for a civil rights violation $\Rightarrow Q_{\min} = 0.25$
- Pattern-or-practice finding $\Rightarrow Q_{\min} = 0.20$

All floor applications are recorded in the score record before the Alignment Score is computed.

Normalization

$$D_{\text{norm}} = \frac{D_{\text{raw}} - D_{\text{min}}}{D_{\text{max}} - D_{\text{min}}}$$

Bounds are configurable per entity class. Values are clamped to $[0, 1]$ — no extrapolation permitted.

Reproducibility

A reproducibility record stores input hashes, configuration versions, code version, and output hash. Any score can be re-derived byte-for-byte from the quantum set and the potential profile version.

Vertical Taxonomy

Alignment.V1 uses one universal evidence and scoring framework across multiple verticals. Vertical profiles describe differences in mandate, legal duties, peer groups, data availability, and valid metrics; they do not create separate scoring engines.

Top-Level Families

1. **Legislative:** legislatures, councils, committees, and related public oversight bodies.
2. **Executive:** departments, agencies, regulators, public administrators, health authorities, human services, and corrections.
3. **Judicial:** judges, courts, clerks, prosecutors, and other adjudicative or justice-system entities.
4. **Consumer-facing and commercial:** credit reporting agencies, lenders, insurers, employers, platforms, utilities, and other entities whose conduct directly affects consumers.

These families are an organizational taxonomy, not a presumption that every member has identical metrics. A Department of Corrections and a health authority are both Executive entities but require different potential profiles.

First Validation Verticals

The first cross-vertical validation set is deliberately heterogeneous:

Family	Example profile	Primary validation purpose
Executive	Department of Corrections	custody, safety, due process, rehabilitation, access, and accountability
Consumer-facing	Nationwide Consumer Reporting Agency	data accuracy, dispute handling, transparency, equity, and remediation
Judicial	Oregon circuit court or judge	procedural consistency, timeliness, access, and reasoned decision-making
Executive	Oregon Health Authority	public health administration, access, outcomes, and data stewardship

Equifax is the first planned live entity instance, but the NCRA profile is the reusable vertical artifact. A corrections department is the second validation profile, not an Equifax-specific extension.

Vertical Profile Contract

A vertical profile defines how a class of entities expresses institutional potential. It supplies configuration to the universal engine without changing the evidence, provenance, or score-result contracts.

Profile Shape

```
{
  "profile_id": "ncra-v1",
  "vertical_family": "consumer_facing",
  "entity_class": "nationwide_consumer_reporting_agency",
  "legal_regimes": ["FCRA", "Regulation V"],
  "peer_class": { "id": "ncra-primary", "minimum_size": 3 },
  "dimensions": {
    "F": { "metrics": ["LMCR", "PAS", "RVR", "SOF"] },
    "C": { "metrics": ["OVI", "PAR", "CDS", "DRR"] },
    "T": { "metrics": ["RDR", "RTI", "FCR", "PAX"] },
    "E": { "metrics": ["CRR", "PTS", "RUI", "BCR"] },
    "Q": { "metrics": ["DDI", "GAS", "PCOR", "ASR"] },
    "Ac": { "metrics": ["SCR", "AC", "CRR_c", "RAR"] }
  },
  "normalization_version": "ncra-norm-v1",
  "sufficiency_rules": {},
  "hard_floors": {},
  "caps": {}
}
```

A corrections profile may use different metric codes, sources, peer classes, and thresholds while retaining the same six-dimensional result contract. The engine must reject incomplete profiles rather than silently substituting metrics from another vertical.

Entity and Activity Separation

An entity profile identifies the subject; an activity record identifies what that subject does. This distinction is mandatory for corporate groups and public departments with multiple programs.

```
entity -> legal entity -> activity -> jurisdiction -> legal regime
      -> evidence
```

A label such as “debt collector” must therefore be attached only when the source supports the named legal entity, activity, jurisdiction, and applicable period. It must not be generalized across a corporate group without evidence.

Nationwide Consumer Reporting Agency Profile

This section incorporates the detailed Beacon Scoring Engine baseline for the Nationwide Consumer Reporting Agency (NCRA) class. It is a vertical profile, not a special scoring engine for Equifax.

Class Definition

Institution class	Nationwide Consumer Reporting Agency (NCRA)
Regulatory framework	FCRA, 15 U.S.C.
S 1681 et seq.; Regulation	
V, 12 C.F.R. Part 1022	
Primary regulator	Consumer Financial Protection Bureau
Secondary regulators	FTC, state regulators, and state attorneys general
Primary peer class	Three nationwide CRAs; specialty CRAs are a secondary cohort
Scoring cadence	Quarterly dimensions; annual Alignment Score, subject to evidence sufficiency

Metric Registry

The NCRA profile uses the Beacon metric registry:

Dimension	Metrics
Fidelity	LMCR, PAS, RVR, SOF
Consistency	OVI, PAR, CDS, DRR
Transparency	RDR, RTI, FCR, PAX
Efficiency	CRR, PTS, RUI, BCR
Equity	DDI, GAS, PCOR, ASR
Accountability	SCR, AC, CRR _c , RAR

Each metric retains the Beacon specification’s definition, weight, source requirements, direction, normalization bounds, minimum sample rules, flags, and hard-floor or cap behavior. These must be represented in the executable metric registry before an empirical score is published.

Complete Evidence Inventory

The NCRA intake should be capable of cataloging, without treating every item as score-eligible:

- CFPB enforcement actions, consent orders, complaint data, and public research;
- FTC actions, state attorney general actions, and court decisions;
- SEC filings, annual reports, quarterly filings, legal proceedings, risk disclosures, and cyber-security disclosures;
- FCRA litigation, class settlements, appellate decisions, and monitor reports;
- dispute timelines, reinvestigation outcomes, furnisher notifications, security freezes, and consumer disclosure records;
- demographic, geographic, language-access, thin-file, and no-file research; and
- remediation, audit, consent-order monitoring, and recidivism records.

Complaint records are observations about reported consumer experiences. They are not, by themselves, adjudicated findings of wrongdoing and cannot be converted directly into a negative Alignment Score component.

Equifax Pilot Scope

The first live instance is Equifax as a corporate group. The profile must separately resolve Equifax Inc., named subsidiaries, business segments, activities, jurisdictions, legal regimes, and effective periods. Debt collection or recovery-management activity is recorded only where a source supports that scoped classification. The pilot must preserve uncertainty rather than assigning the label to every Equifax entity.

The first output is an auditable evidence and readiness profile. A final Alignment Score is emitted only when the NCRA profile's dimensional evidence requirements are satisfied.

Department of Corrections Profile

The Department of Corrections (DOC) profile is the second validation vertical. It demonstrates that the universal engine can evaluate a public custodial institution without importing consumer-credit metrics.

Profile Scope

The DOC profile may cover a state department, facility, division, or program, but the subject and period must be explicit. Facility-level evidence must not be generalized to the entire department without a documented aggregation rule.

Candidate Metric Families

Dimension	Candidate DOC measures
Fidelity	compliance with constitutional, statutory, court, and department mandates
Consistency	comparable disciplinary, classification, grievance, and release decisions
Transparency	public reporting, records access, incident disclosure, and grievance visibility
Efficiency	case processing, staffing capacity, program delivery, and backlog measures
Equity	demographic disparity, disability access, language access, and geographic/facility differences
Accountability	corrective actions, inspections, investigations, remediation, and recurrence

These are candidate families, not approved metrics. Each metric requires a written definition, valid denominator, source inventory, normalization rule, sufficiency threshold, and legal review before scoring.

DOC-Specific Safeguards

- Separate department, facility, contractor, and individual officer entities.
- Distinguish allegations, grievances, investigations, findings, and adjudicated outcomes.
- Never use incarceration or demographic outcome differences without appropriate denominators and documented confounder handling.
- Treat missing or suppressed correctional data as a transparency and evidence-availability question, not automatic proof of misconduct.
- Preserve facility-level and population-level results separately.

The DOC profile is successful only if it runs through the same universal entity, evidence, metric-observation, and score-result contracts as the NCRA profile while producing a distinct, profile-versioned metric registry.

Executable Metric Registry and Cross-Vertical Validation

The detailed Beacon sub-metrics are being converted into versioned executable registries. The registry is the bridge between the normative specification and the scoring engine. It records each metric's dimension, weight, direction, required source types, minimum sample size where applicable, and profile version.

```
metric code -> dimension -> formula -> source requirements
           -> normalization -> sufficiency -> flags/floors
```

The first reusable registries are:

- `ncra-v1`: the 24-metric Nationwide Consumer Reporting Agency registry derived from the Beacon baseline;
- `corrections-v1`: an independent Department of Corrections registry with custody, due process, grievance, access, capacity, and remediation measures.

Both profiles use the same universal entity, evidence, and score-result contracts. Their metric codes, source requirements, peer classes, and interpretive rules differ by design. Equifax is an entity instance of `ncra-v1`; it is not the profile itself.

At this phase, registry validation and profile separation are implemented. Metric-specific formulas, live source adapters, raw snapshot persistence, and empirical calibration remain subsequent implementation phases. No registry fixture alone constitutes a real-world Alignment Score.

Metric Observation Layer

The metric observation layer is the next executable bridge between source records and Alignment Scores. It is deliberately vertical-neutral: an observation identifies the entity, metric, raw value, normalization bounds, source, citation, date, jurisdiction, sample size, and confidence. The vertical registry supplies the metric direction, weight, and minimum sample requirements.

```
raw source record
-> metric observation with provenance
-> bounded normalization (inverse metrics are reversed)
-> sample sufficiency check
-> confidence-weighted metric aggregate
-> registry-weighted dimension value
-> universal Alignment Score contract
```

The same machinery can process an NCRA observation such as LMCR and a corrections observation such as DPA; only the vertical registry, source mapping, and evidence interpretation differ. A first use case does not become a system-wide assumption.

Observations that do not meet a metric's minimum sample requirement are retained for audit and shown as insufficient, but are excluded from the metric aggregate. Every observation retains source provenance so that a score can be reproduced from the underlying record set.

This layer does not yet fetch live sources or persist snapshots. Those remain separate ingestion and storage concerns and must be completed before an empirical profile is published.

Immutable Source Snapshot Layer

Before a source record is transformed into a metric observation, the system preserves the retrieved response as an immutable source snapshot. This is a vertical-neutral storage contract: the same structure can contain an SEC filing for an NCRA, an inspection report for a corrections department, or a court decision for a judiciary profile.

```
entity_id + source_type + source_record_id
-> retrieved payload
-> canonical content hash
-> immutable snapshot
-> parser/versioned observation
```

Each snapshot records the entity, source type, source identifier, retrieval time, optional source URL, parser version, payload, and content hash. The content hash permits integrity verification and prevents a later source change from silently altering a previously computed result. A parser may produce a new observation version, but it must not mutate the original snapshot.

Snapshot integrity is checked before selection for downstream processing. Invalid or tampered snapshots are excluded rather than silently repaired. This establishes the reproducibility boundary required by the Beacon baseline: auditors can re-run a parser against the preserved source response and compare both the source hash and the resulting observation.

The current implementation provides deterministic in-memory contracts. Durable object storage, retention policy, encrypted storage for restricted material, source-specific fetch adapters, and signed external manifests remain later infrastructure work. No snapshot by itself constitutes a validated metric or published Alignment Score.

Governance and Reproducibility

- Public institutional records are the primary and sufficient evidentiary basis; individual submissions are supplementary.
- Every claim and score component carries a citation chain.
- Raw evidence is immutable; derived outputs are append-only revisions.
- Hard floors are applied and logged before the Alignment Score is computed.
- A reproducibility record stores input hashes, configuration versions, code version, and output hash.
- No Alignment Score is a finding of individual wrongdoing, a prediction of future conduct, or a substitute for an official finding.

Implementation Phases

The following phases are tracked in the repository as they are built. The current implementation status is recorded at the top of each phase.

1. Shared evidence foundation and provenance contracts (built).
2. Universal Alignment Score engine with insufficient-data and hard floors (built).
3. Judiciary filter experience with case-type filtering and the scale-and-rotate panel (built).
4. Identity system: gold A mark, hero first letter, favicon (built).
5. Universal metric registry and vertical profile contract (built).
6. NCRA profile and Equifax role-aware intake (specified; first live instance).
7. Department of Corrections profile and cross-vertical fixtures (specified; reproducibility validation).
8. Registry-driven metric observation layer (built).

9. Source adapters, raw snapshots, and catalog discovery (specified).
10. Normalization service with per-class bounds (specified).
11. Public score API (specified).
12. Reproducibility audit tooling (specified).